Exercise 7: Text Generation Using Google Gemini LLM API

Aim

To develop a Python program using the Google Gemini Large Language Model (LLM) API to generate text responses for different user prompts and display the generated output.

Algorithm

1. Install the Google Gemini Python SDK.
2. Obtain an API key from Google AI Studio.
3. Import the required Python library.
4. Create a Gemini API client using the API key.
5. Get a prompt from the user.
6. Send the prompt to the Gemini model.
7. Display the generated text response.

PROGRAM

```
from google import genai


api_key = input("Enter your Gemini API key: ")

client = genai.Client(api_key=api_key)


while True:
```

```
    prompt = input("\nEnter your prompt (or 'exit' to stop): ")
    if prompt.lower() == "exit":
        print("Program terminated.")
        break
    try:
        response = client.models.generate_content(
            model="gemini3.8-flash",
            contents=prompt
        )
        print("\nGenerated Output:")
        print(response.text)
    except Exception as e:
        print("Error:", e)
```

SAMPLE OUTPUT

```
...  Enter your Gemini API key: AQ.Ab8RN6IQ1BY3EKyyeWQD87r2gVvrhO-HqAcezGbm8QCTViFE9Q

     Enter your prompt (or 'exit' to stop): Artificial Intelligence (AI) is a technology that enables comput

     Generated Output:
     That is a clear and accurate summary of Artificial Intelligence (AI).

     To expand on the fields you mentioned, here are a few specific ways AI is being applied today:

     *   **Healthcare:** AI assists in analyzing medical imaging (like X-rays and MRIs) for early disease de
     *   **Education:** It powers adaptive learning platforms that adjust to a student's individual pace, au
     *   **Transportation:** Beyond self-driving cars and drones, AI optimizes city traffic flow, plans effi
     *   **Finance:** Algorithms detect fraudulent transactions in real-time, assess credit risk, and manage
     *   **Customer Service & Creative Industries:** Generative AI tools assist in writing, graphic design,

     Would you like to explore how a specific type of AI works (like Machine Learning or Deep Learning), div
```

RESULT

Thus, the Python program was developed to accept different user prompts, send them to the Google Gemini LLM API, and display the generated text responses successfully.